

A finite-math boundary-element modeling workflow for OCD deep TMS

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ABSTRACT

We describe a finite, reproducible visualization workflow for an obsessive-compulsive disorder (OCD) tab in a neuromodulation research interface. The workflow combines a bounded frequency-intensity grid, a single-layer boundary-element (BEM) cortical potential calculation, layered attenuation, and illustrative session-indexed response curves. All results are outputs of a simplified computational model. They are not patient-specific field estimates, clinical recommendations, evidence of efficacy, or a substitute for regulatory-cleared planning systems.

Keywords: boundary element method; finite grid optimization; deep TMS visualization; OCD research; computational modeling

1. Model scope

The model intentionally constrains the OCD scenario to a finite 18-22 Hz by 80-120% MSO parameter rectangle. Each lattice point is reduced by a modular stability function and evaluated as a visualization aid. The cortical mesh is a deformed spherical boundary under a dipole source; it is not an anatomical head model and does not encode a patient MRI, coil registration, stimulation threshold, or treatment response.

2. Finite mathematical formulation

$$\text{phi}_i = (\mu_0 / 4\pi) [\mathbf{m} \cdot (\mathbf{x}_i - \mathbf{x}_{\text{coil}})] / \max(\|\mathbf{x}_i - \mathbf{x}_{\text{coil}}\|^3, \epsilon), i = 1, \dots, N$$

Equation 1 | Discrete single-layer BEM potential evaluated at N boundary nodes.

$$E(d_j) = I_0 \text{product}_{\ell: d_j \geq b_{\ell\ell}} \exp[-4 \sigma_{\ell\ell} (d_j - b_{\ell\ell})], j = 1, \dots, 50$$

Equation 2 | Finite layered attenuation through the configured conductivity layers.

$$(f^*, I^*) = \operatorname{argmax}_{\{f \in F, I \in J\}} \Phi(R(f + iI/100)), |F| = |J| = 46$$

Equation 3 | Finite-grid selection of the modular stability maximum after reduction R.

$$Z_{pq} = E_{\max} [\exp(-a r_{\text{left},pq}) + \exp(-a r_{\text{right},pq})] [1 + 0.15 \cos(2X_{pq}) \sin(Y_{pq})]$$

Equation 4 | Finite dual-lobe field visualization over the tab's 50 x 50 mesh.

$$s_k = \max(8, 34 - 0.85k + \eta_k), k = 1, \dots, 29$$

Equation 5 | Illustrative finite-session severity series; η_k is synthetic perturbation.

3. Data-derived model characteristics

Characteristic	Finite-model result	Interpretation
Sweep domain	18-22 Hz; 80-120% MSO	OCD tab scenario bounds
Grid maximum	18.00 Hz; 100.44%	Discrete stability argmax
Stability score	0.999921	Dimensionless model score
Peak cortical potential	22.0998	Scaled BEM surface quantity
Depth endpoint	0.017% at 45 mm	Layered attenuation output
Finite session series	29 simulated sessions	Illustrative, not clinical outcome

Table 1 | Characteristics computed during this preprint run from the same bounded BEM pathway displayed by the OCD treatment tab.

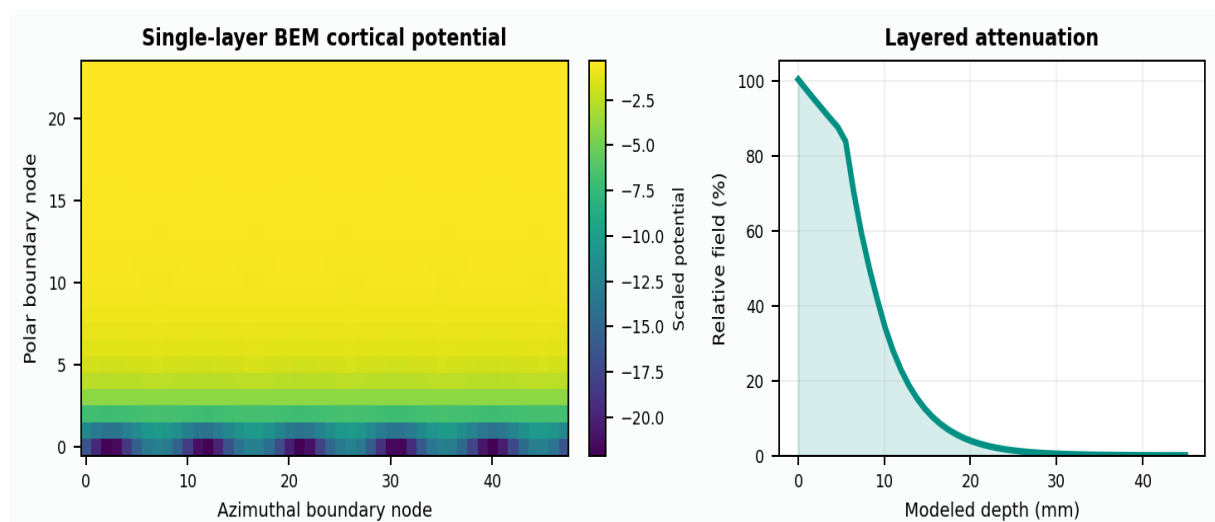


Figure 1 | The left panel visualizes the BEM potential over the discrete cortical boundary mesh. The right panel reports the model's layered depth attenuation. Values are normalized/scaled implementation outputs and are not in-vivo dosimetry.

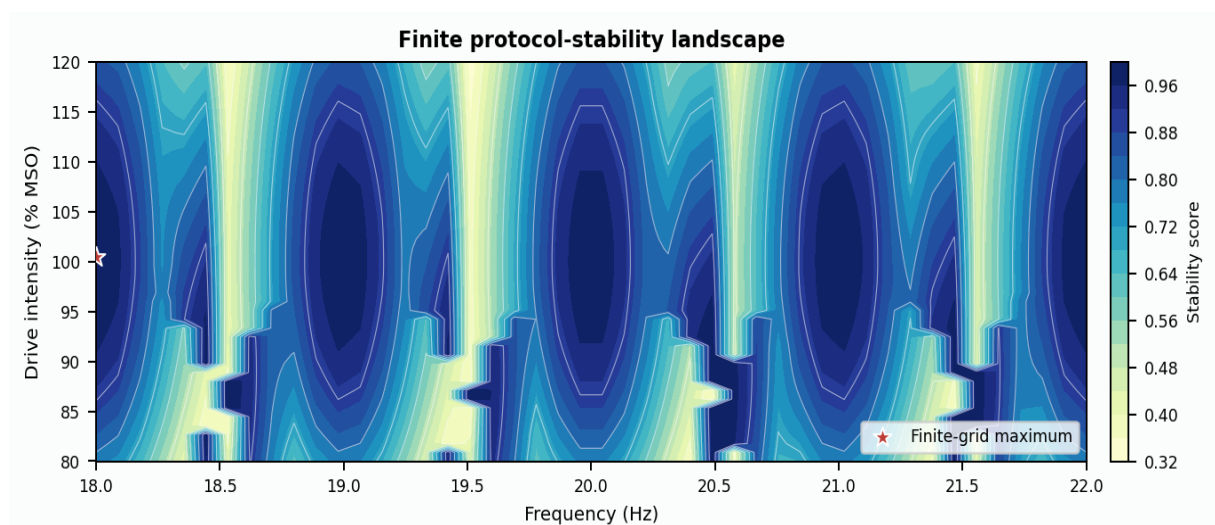


Figure 2 | Bounded stability landscape used by the OCD tab. The star identifies the maximum on the 46 by 46 finite grid, not a patient-level optimum.

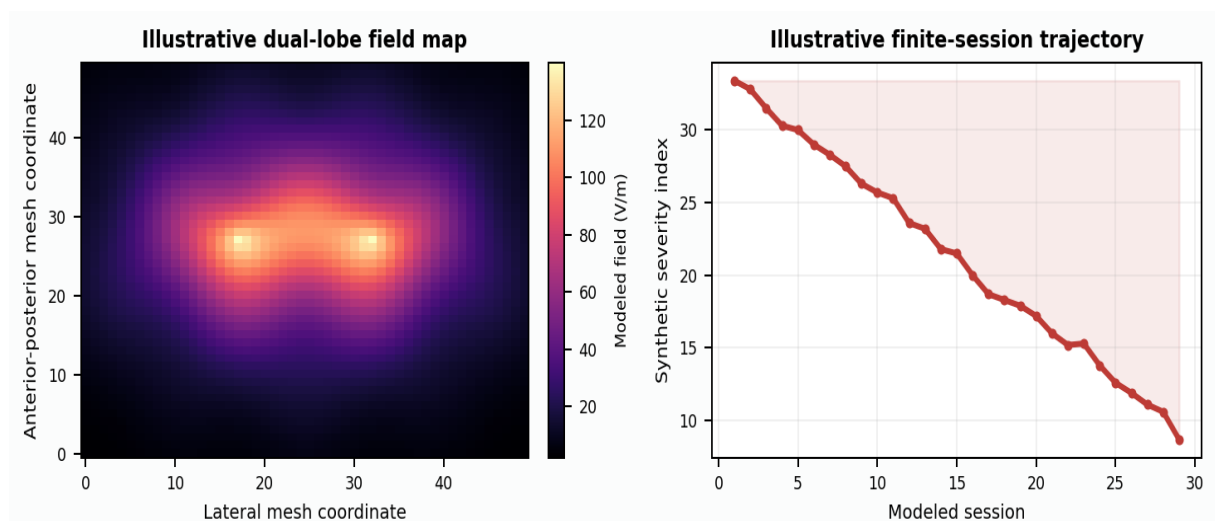


Figure 3 | Existing OCD-tab field visualization and synthetic 29-session series. The session plot is a display model and makes no clinical efficacy claim.

4. Interpretation and limitations

The figures demonstrate that a finite BEM pathway can make its assumptions visible: a source-to-boundary potential, a layered attenuation curve, and a bounded search landscape. The apparent maximum is determined by the defined objective and mesh, so it should be interpreted only as a property of this model. A clinically usable field model would require MRI-derived segmentation, validated tissue conductivities, coil placement and orientation, calibrated stimulator waveforms, uncertainty quantification, and independent validation.

The illustrative response series is purposefully separated from the field calculations. It is synthetic, includes randomly generated perturbations, and has no patient data, comparator arm, p-value, or outcome-validation basis. It must not be used to predict symptom change, choose treatment parameters, or support a clinical decision.

5. Reproducibility

This document was produced from the repository functions `get_oed_fea_simulation()` and `get_moduli_bem_paradigm(condition='oed', freq_range=(18.0, 22.0), intensity_range=(80.0, 120.0))`. The generator records computed values in Table 1 and embeds the resulting plots directly in the PDF. The finite-grid and mesh sizes are part of the source implementation, allowing the visualization to be rerun under the same assumptions.

Data and code availability

The simulation code and PDF generator are included in the local NeuroMorph rTMS workspace. No human participant data were used to create this preprint.